Comprehensive Analysis of Small Molecule Modulators of SARS-CoV-2 Viral Load: Inhibitory Mechanisms, Pro-viral Stimulants, and Computational Discovery Paradigms

The pharmacological management of COVID-19 has necessitated a transition from rapid, empirical drug repurposing to a high-resolution, target-driven strategy mediated by computational informatics and structural biology. Central to this evolution is the identification of small molecules whose interactions with both viral and host proteins are highly correlated with changes in viral load—the primary clinical end point of pharmacological intervention.[1] By leveraging extensive knowledge bases, including those formatted in logical frameworks such as Prolog for relationship mapping, researchers have identified a diverse array of chemical entities that either suppress viral progression or, in specific immunological contexts, paradoxically facilitate it.

The Informatic Architecture of Viral Load Modulation

The identification of molecules correlated with the stimulation and inhibition of COVID-19 viral load relies heavily on the integration of disparate biological datasets. Knowledge graphs, such as the Drug Repurposing Knowledge Graph (DRKG), normalize data from over a dozen entity types, including human genes, chemical compounds, biological processes, and drug side effects, encompassing nearly six million relationships.[2] These platforms utilize machine learning and logic-based algorithms to infer direct and indirect links between drugs and cellular products associated with the SARS-CoV-2 lifecycle.[2]

Within these computational frameworks, drug repurposing is cast as a task of link prediction or knowledge graph completion.[4] Algorithms designed for motif discovery identify high-order patterns of interest, such as specific interaction sequences between a compound, a host protease, and a viral structural protein.[3] These correlations provide the foundation for selecting candidates for clinical evaluation, shifting away from traditional de novo discovery which is often too time-consuming to address an active pandemic.[4]

Mechanisms of Small-Molecule Inhibition of Viral Load

Inhibition of viral load is achieved through the disruption of the three critical phases of the SARS-CoV-2 lifecycle: attachment and entry, intracellular replication and transcription, and assembly and exit. Small molecules targeting these stages are identified as highly correlated

with the reduction of viral titers and improved clinical outcomes.[6]

Blockade of Viral Attachment and Cellular Entry

The interaction between the SARS-CoV-2 spike (S) protein and the human angiotensin-converting enzyme 2 (hACE2) receptor is the initiating event of infection.[8] Small-molecule inhibitors (SMIs) of this protein-protein interaction (PPI) represent a crucial therapeutic class. Unlike monoclonal antibodies, SMIs may offer oral or inhaled administration routes and reduced sensitivity to viral mutations.[8]

Key molecules identified as inhibitors of the S-ACE2 interaction include specific dyes and novel drug-like compounds. Compounds such as DRI-C23041 and DRI-C91005 have demonstrated low micromolar activity in cell-free assays.[8] DRI-C23041, in particular, exhibits selectivity in blocking the entry of SARS-CoV-2 spike-expressing pseudoviruses into hACE2-expressing cells with an IC_{50} of approximately 6–7 μM.[8] Other identified molecules include the dyes Congo red, direct violet 1, and Evans blue, though these are often considered promiscuous inhibitors.[8]

Beyond direct PPI blockade, the entry process is inhibited by molecules targeting host proteases required for "priming" the spike protein. For fusion to occur, the S protein must be cleaved by host enzymes like TMPRSS2 or Cathepsin L.[7]

Target Component	Small Molecule Inhibitor	Mechanism of Inhibition
S-ACE2 PPI	DRI-C23041	Competitive or steric blockade of the receptor-binding domain (RBD).[8]
TMPRSS2	Camostat mesylate	Inhibition of the serine protease that cleaves the S2' site at the cell surface.[7]
TMPRSS2	Nafamostat mesylate	High-potency inhibition of surface proteolytic priming.[7]
Cathepsin L	K777 / E64-d	Inhibition of lysosomal proteases involved in endosomal viral entry.[11]
Furin	MI-1851	Prevention of pre-activation cleavage during viral

		maturation.[11]
PIKfyve Kinase	Apilimod (STA-5326)	Disruption of endosomal trafficking and membrane fusion.[7]

The efficiency of these molecules is highly correlated with the specific entry route utilized by the virus in different tissues. For example, while chloroquine and hydroxychloroquine were early candidates due to their ability to raise endosomal pH and inhibit cathepsins, their clinical efficacy in the lungs was limited because the virus primarily utilizes the TMPRSS2-mediated surface entry route in pulmonary cells.[7]

Inhibition of the Viral Protease Complex

Following entry and uncoating, the SARS-CoV-2 genome is translated into polyproteins pp1a and pp1ab, which must be cleaved by viral cysteine proteases to yield 16 functional non-structural proteins (nsps).[7] The primary targets in this phase are the main protease (M^{pro}, also known as $3CL^{pro}$ or nsp5) and the papain-like protease (PL^{pro}, nsp3).

The main protease is a homodimeric enzyme containing a catalytic dyad of His41 and Cys145.[13] Inhibition of M^{pro} is highly correlated with a reduction in viral load because it effectively shuts down the formation of the replicase-transcriptase complex (RTC).[16] Nirmatrelvir (the active component of Paxlovid) is the most prominent clinical inhibitor, acting as a covalent binder to the active site.[6] Other potent molecules identified through molecular dynamics simulations and crystallography include α-ketoamides and ABBV-744, with the latter showing a strong binding affinity (ΔG_{bind} of -45.43 $kcal/mol$).[14]

Molecule Identifier	Protease Target	Binding Characteristics	Mechanism of Action
Nirmatrelvir (PF-07321332)	M^{pro} (3CLpro)	Covalent, Peptidomimetic	Blocks cleavage of pp1a/pp1ab at 11 sites.[6]
PF-00835231	M^{pro} (3CLpro)	Covalent, Ketone-based	Active form of prodrug PF-07304814; high identity across CoVs.[16]

GC-376	M^{pro} (3CLpro)	Pre-clinical inhibitor	Wide-spectrum inhibition of coronaviridae proteases.[17]
ABBV-744	M^{pro} (3CLpro)	Strong Affinity	Repurposed lead compound identified via pharmacophore modeling.[18]
Nicotiflorin	PL^{pro} / M^{pro}	Phytochemical	Low binding energy (-7.40 $kcal/mol$) to PL^{pro}.[19]
Caffeoylshikimic acid	M^{pro} (3CLpro)	Phytochemical	High affinity identified in *Hibiscus sabdariffa* extracts.[20]

Analysis of the catalytic pockets reveals that the potencies of these inhibitors are determined by the specific interplay with the His41 residue, a component of the catalytic dyad that is often overlooked in favor of the Cys145 residue.[15] Dual-acting inhibitors targeting both M^{pro} and PL^{pro} are currently under investigation to prevent the emergence of resistant viral strains.[21]

Disruption of the RNA-Dependent RNA Polymerase (RdRp)

The RdRp complex is the central engine for viral genome replication and transcription. Small molecules that function as nucleoside analogues are highly correlated with the inhibition of viral load by inducing premature chain termination or lethal mutations.[6]

Remdesivir ($GS-5734$) is the most established RdRp inhibitor. As a monophosphoramidate prodrug, it enters cells and is metabolized to the active triphosphate form, which competes with ATP for incorporation into nascent RNA.[6] This incorporation causes delayed chain termination at position $i+3$, effectively stopping the replication of the 30 kb genome.[16] Molnupiravir (EIDD-2801) offers a different mechanism; as a ribonucleoside analog, it induces "error catastrophe." By alternating between forms that mimic cytidine and uridine, it introduces widespread mutations into the viral genome during replication, leading to non-functional viral progeny.[6]

RdRp Inhibitor	Form	Mechanism of Effect	Status
Remdesivir	IV Injection	Delayed RNA chain termination.[6]	FDA Approved
Molnupiravir	Oral Tablet	Induction of viral error catastrophe.[6]	FDA EUA
Favipiravir	Oral Tablet	Purine analog, RdRp inhibition/mutagenesis.[6]	Used in select regions
AT-527	Oral Tablet	niRAN (Nidovirus RdRp-associated nucleotidyltransferase) antagonist.[6]	Clinical Trial

A critical insight into the correlation between these molecules and viral load is the importance of "intracellular bioavailability." Mathematical models of infection dynamics suggest that the in vivo antiviral activity of remdesivir and favipiravir is directly proportional to their concentration in alveolar type 2 cells.[23] If lung concentrations remain below the predicted EC_{50}, viral replication proceeds largely unchecked, emphasizing the need for therapies that achieve high tropism for the primary site of infection.[23]

Pro-viral Small Molecules and Stimulators of Viral Load

A nuanced understanding of COVID-19 pathogenesis involves identifying molecules that, while potentially beneficial for treating symptoms, are positively correlated with increased viral load or spread. These "pro-viral" effects often arise from the suppression of innate antiviral defenses or the modulation of host metabolism in a manner that favors the virus.

The Janus Kinase (JAK) Inhibitor Paradigm

JAK inhibitors such as baricitinib, ruxolitinib, and tofacitinib are widely used to treat the hyperinflammatory state or "cytokine storm" associated with severe COVID-19 by blocking the signaling of cytokines like IL-6.[22] However, these molecules are also highly correlated with the stimulation of viral replication when administered in certain contexts.[26]

The mechanism for this pro-viral effect centers on the disruption of the Type I and Type III interferon (IFN) response. Under normal conditions, infected cells secrete IFNs that bind to receptors on neighboring uninfected cells, activating the JAK/STAT signaling pathway.[9] This activation triggers the transcription of interferon-stimulated genes (ISGs), which establish an

antiviral state in the tissue.[26] By inhibiting JAK1 and JAK2, molecules like baricitinib remove these innate barriers, enabling the virus to infect previously protected cells and amplify its viral load.[26]

JAK Inhibitor	Target	Potential Pro-viral Consequence	Clinical Correlation
Baricitinib	JAK1/JAK2	Suppression of ISG production; increased viral spread.[26]	Correlated with adverse outcomes if used prematurely.[26]
Ruxolitinib	JAK1/JAK2	Removal of IFN-mediated barriers to infection.[26]	Investigated for cytokine storm management.[7]
Tofacitinib	JAK1/JAK3	Disruption of mucosal antiviral immunity.[26]	Used in hospitalized patients.[24]

This pro-viral effect is a function of the drug concentration and the timing of therapy. Early administration of JAK inhibitors may exacerbate infection, whereas late administration may be necessary to prevent organ failure from the host's own immune response.[1]

Metabolic and Oxidative Stress Modulators

Viruses frequently hijack host metabolic pathways to optimize the production of viral components. Evidence from related respiratory viruses, such as influenza, suggests that the down-modulation of glucose-6-phosphate dehydrogenase (G6PD) expression is highly correlated with increased viral replication.[27] G6PD is responsible for producing NADPH, which regenerates reduced glutathione (GSH) to maintain redox homeostasis. In G6PD-deficient environments, elevated reactive oxygen species (ROS) levels favor viral proliferation.[27] Molecules that act as SIRT2 activators can rescue GSH production and are negatively correlated with viral load, whereas SIRT2 inhibitors or agents that increase oxidative stress act as pro-viral stimulators.[27]

Host Factors and Thromboinflammatory Correlates

The severity of COVID-19 and the magnitude of the viral load are positively correlated with specific host molecules involved in coagulation and inflammation. Hyperactive platelets act as pro-viral factors by facilitating the transport of viral particles and increasing the systemic

inflammatory state.[28]

Key molecules in this category include:

- **P-Selectin:** Elevated surface expression on platelets is correlated with the formation of platelet-monocyte aggregates and is a predictor of thrombosis in severe COVID-19.[28]
- **Von Willebrand Factor (VWF):** High plasma concentrations are strongly correlated with mortality and disease severity, reflecting the endotheliopathy caused by the virus.[28]
- **Interleukin-6 (IL-6):** While primarily a marker of the immune response, its levels are directly related to the hyperinflammatory subphenotype and the "cytokine storm" that follows rampant viral replication.[24]

Host-Targeted Small Molecules and Lysosomotropic Agents

Host-targeted therapies offer the advantage of broad-spectrum activity and a higher barrier to resistance. Several classes of small molecules have been identified as negatively correlated with viral load through the modulation of host cell pathways.

Kinase and Cytoskeletal Inhibitors

Small molecules targeting host kinases involved in the viral lifecycle have shown significant inhibitory potential. Bemcentinib, an AXL kinase inhibitor, blocks viral entry via cell membrane binding and fusion.[6] Silmitasertib targets casein kinase 2 (CK2), which is exploited by the virus for its assembly and replication.[6] Additionally, the disruption of the host cell's transport machinery has proven effective. Sabizabulin, an inhibitor of α- and β-tubulin, disrupts the cytoskeleton, thereby preventing the intracellular movement and eventual release of newly formed virions.[6]

Lysosomotropic Drugs and Endolysosomal Signaling

A growing body of research identifies the endolysosome as a primary target for COVID-19 prophylaxis and treatment.[24] Small molecules with lysosomotropic characteristics—those that accumulate in the acidic compartments of the cell—interfere with the pH-dependent processes required for viral entry and exit.[24]

Lysosomotropic Molecule	Pharmacological Class	Role in COVID-19
Fluvoxamine	SSRI	Demonstrated clinical benefit in reducing hospitalization.[24]

Amitriptyline	Tricyclic Antidepressant	Interference with lysosome-dependent viral signaling.[24]
Desloratadine	Antihistamine	Modulation of pro-inflammatory mediators and viral entry.[24]
Azelastine	Antihistamine	Investigated for multi-target approach in sepsis/COVID-19.[24]
Ambroxol	Secretolytic agent	Clinical benefits observed in rodent models of sepsis.[24]

These drugs offer a multi-target approach by addressing both the viral infection of host cells and the shedding of toll-like receptors (TLRs), thereby tempering the expression of pro-inflammatory mediators such as TNF-α, IL-1β, and IL-6.[24] Rational drug repurposing of these approved molecules represents a major opportunity to improve outcomes, especially in settings where access to specialized antivirals like remdesivir is limited.

Natural Product Discovery via Knowledge Graph Link Prediction

The identification of natural products as inhibitors of SARS-CoV-2 has been facilitated by high-throughput screening and knowledge graph completion. Phytochemicals derived from various plants have shown high binding affinities for viral proteases and structural proteins.

Phytochemical Inhibitors of 3CLpro and PLpro

Molecular docking studies on phytochemicals from *Hibiscus sabdariffa* have identified several molecules with significant inhibitory potential. Compounds such as caffeoylshikimic acid and nicotiflorin exhibit low binding energies to M^{pro} and PL^{pro}.[19] Caffeoylshikimic acid, in particular, was found to have a binding affinity of -7.72 $kcal/mol$ to M^{pro}, making it a promising candidate for further in vitro validation.[19]

Another notable discovery is Cambogin, a natural product from the *Garcinia* genus. Computational analysis identified Cambogin as a modulator of 46 host genes affected by both SARS-CoV-2 and KSHV infections.[30] Cambogin not only inhibits the replication and production of SARS-CoV-2 virions in vivo by reducing the expression of EGFR and Cyclin A2 but also

demonstrates efficacy in murine models of co-infection.[30]

Volatile Compounds and Inhalants

In pursuit of therapeutic options that can be administered directly to the respiratory tract, volatile molecules from natural medicinal inhalants like turmeric (*Curcuma longa*) have been screened.[31] Seven chemicals from turmeric have been identified as adhering to drug-likeness limits (Lipinski's Rule of Five) and interacting favorably with the spike-associated domains and the main protease ($6LU7$ and $6M0J$).[31] Two active molecules from this group were predicted to have exceptionally strong binding affinities, suggesting their potential as prophylactic or therapeutic inhalants.[31]

Correlates of Protection and Viral Load Kinetics

To evaluate the success of small-molecule interventions, it is necessary to identify measurable correlates that track with the viral load. The cycle threshold (Ct) value in RT-PCR is the most widely used indirect indicator of viral load, where Ct is inversely proportional to the amount of viral RNA present.[32]

Biochemical Correlates of High Viral Load

Infected individuals exhibit a range of biochemical derangements that are highly correlated with their viral load and disease severity.[32] These markers are essential for prognostic monitoring and for evaluating the effectiveness of small-molecule inhibitors in clinical trials.

Biochemical Parameter	Correlation with Viral Load	Clinical Context
ALT / AST	Positive	Markers of systemic inflammation and liver involvement.[32]
C-Reactive Protein (CRP)	High Positive	Often increased; very high levels predict unfavorable outcomes.[29]
Ferritin	Positive	Correlated with the hyperinflammatory subphenotype.[24]
D-dimer	High Positive	Elevated in severe cases; marker of

		hypercoagulability.[29]
Lactate Dehydrogenase (LDH)	Positive	Values >400 IU/ml associated with severe disease and ARDS.[29]
Total Leucocyte Count (TLC)	Positive	Neutrophilia and lymphopenia are common markers of severity.[29]

Research indicates that the severity of COVID-19, particularly the progression to Acute Respiratory Distress Syndrome (ARDS), is characterized by high levels of interleukins and low blood pressure, belonging to a "hyperinflammatory subphenotype".[29] Small molecules targeting the pathways leading to these biomarkers, such as simvastatin for PEEP management or sodium bicarbonate for acidosis, are used in conjunction with direct antivirals to manage the systemic effects of the infection.[29]

Dynamics of Innate and Adaptive Immunity

The relationship between viral load and the immune response is non-linear. In mild patients, a transient early surge of IFN-β and IP10/CXCL10 is positively correlated with the initial viral load but ultimately predicts later antiviral immunity and successful clearance of the virus.[33] This "coordinated response" of CD4+ and CD8+ T cells and antibody-secreting cells is unique to those who avoid severe disease.[33] Small molecules that enhance these pathways—such as interferon inducers or modulators of "trained immunity"—could be vital in supporting the transition from high viral load to long-term protection.[9]

Synthesis of Findings and Future Directions

The study of small molecules related to COVID-19 viral load has transitioned from a desperate search for repurposed drugs to a sophisticated science of "pharmacological mapping." By combining structural biology (identifying the H41-C145 dyad of M^{pro}) with informatic platforms (like the DRKG and its Prolog-derived logic), researchers have categorized molecules based on their correlation with viral inhibition and stimulation.

The current therapeutic landscape is defined by:

1. **Direct-Acting Antivirals:** Remdesivir, molnupiravir, and nirmatrelvir, which achieve significant reductions in viral load by targeting the core replication machinery.
2. **Host-Targeted Inhibitors:** A new generation of kinase (Bemcentinib, Silmitasertib) and lysosomotropic (Fluvoxamine, Amitriptyline) drugs that offer multi-target protection and are less prone to viral resistance.
3. **The Pro-viral Risk:** A recognition that potent immunomodulators like baricitinib carry a

risk of stimulating viral spread if used in the early, IFN-sensitive phase of infection.

4. **Computational Synergy:** The use of knowledge graphs to discover "hidden linkages," such as the potential for paclitaxel to mitigate cytokine storms or the use of turmeric-derived inhalants.

As the virus continues to mutate, primarily at the spike protein surface, the value of small molecules targeting highly conserved enzymes like $3CL^{pro}$ and RdRp remains paramount. The next phase of research will likely focus on "dual-acting" small molecules and combinatorial therapies that manage the viral load while simultaneously dampening the overexuberant inflammatory response. This multi-pronged approach, supported by the predictive power of logic-based knowledge graphs, remains the most promising path toward the ultimate control of the SARS-CoV-2 pandemic.

Fuentes citadas

1. Innate Immunity Plays a Key Role in Controlling Viral Load in COVID-19: Mechanistic Insights from a Whole-Body Infection Dynamics Model - ACS Publications, acceso: abril 26, 2026, https://pubs.acs.org/doi/10.1021/acsptsci.0c00183
2. Amazon Web Services open-sources biological knowledge graph to fight COVID-19, acceso: abril 26, 2026, https://www.amazon.science/blog/amazon-web-services-open-sources-biological-knowledge-graph-to-fight-covid-19
3. Drug Repurposing for the Treatment of COVID-19: A Knowledge Graph Approach - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8212091/
4. Drug repurposing for COVID-19 via knowledge graph completion ..., acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7869625/
5. Knowledge graph-based approaches to drug repurposing for COVID-19 - CAS, acceso: abril 26, 2026, https://www.cas.org/resources/cas-insights/knowledge-graph-drug-repurposing
6. Emerging small molecule antivirals may fit neatly into COVID-19 ..., acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8882464/
7. Recent advances in small-molecular therapeutics for COVID-19 ..., acceso: abril 26, 2026, https://academic.oup.com/pcm/article/5/4/pbac024/6713978
8. Small-Molecule Inhibitors of the Coronavirus Spike: ACE2 Protein–Protein Interaction as Blockers of Viral Attachment and Entry for SARS-CoV-2 | ACS Infectious Diseases - ACS Publications, acceso: abril 26, 2026, https://pubs.acs.org/doi/10.1021/acsinfecdis.1c00070
9. COVID-19: Coronavirus replication, pathogenesis, and therapeutic strategies, acceso: abril 26, 2026, https://www.ccjm.org/content/87/6/321
10. Small molecules in the treatment of COVID-19 - PMC - NIH, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9719906/
11. Molecular mechanisms of SARS-CoV-2 entry: implications for ..., acceso: abril 26, 2026, https://journals.asm.org/doi/10.1128/mmbr.00260-24

12. Insights into SARS-CoV-2: Small-Molecule Hybrids for COVID-19 Treatment - MDPI, acceso: abril 26, 2026, https://www.mdpi.com/1420-3049/29/22/5403
13. Structural Similarity of SARS-CoV2 Mpro and HCV NS3/4A Proteases Suggests New Approaches for Identifying Existing Drugs Useful as COVID-19 Therapeutics - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7263768/
14. Elucidating Biophysical Basis of Binding of Inhibitors to SARS-CoV-2 Main Protease by Using Molecular Dynamics - ChemRxiv, acceso: abril 26, 2026, https://chemrxiv.org/doi/pdf/10.26434/chemrxiv.12084207.v1
15. Discovery of SARS-CoV-2 3CLPro Peptidomimetic Inhibitors through the Catalytic Dyad Histidine-Specific Protein–Ligand Interactions - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8878928/
16. A Comparative Analysis of SARS-CoV-2 Antivirals Characterizes 3CLpro Inhibitor PF-00835231 as a Potential New Treatment for COVID-19 - ASM Journals, acceso: abril 26, 2026, https://journals.asm.org/doi/10.1128/jvi.01819-20
17. A comparative analysis of SARS-CoV-2 antivirals in human airway models characterizes 3CLpro inhibitor PF-00835231 as a potential new treatment for COVID-19 - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7457613/
18. ABBV-744 as a potential inhibitor of SARS-CoV-2 main protease enzyme against COVID-19, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC7794216/
19. Demonstration of Interaction Between Carbapenem Group Antibiotics and Different Immunosuppressant Drugs by Molecular Docking - ResearchGate, acceso: abril 26, 2026, https://www.researchgate.net/publication/368796455_Demonstration_of_Interaction_Between_Carbapenem_Group_Antibiotics_and_Different_Immunosuppressant_Drugs_by_Molecular_Docking
20. Phytochemicals of Hibiscus sabdariffa with Therapeutic Potential against SARS-CoV-2: A Molecular Docking Study - ResearchGate, acceso: abril 26, 2026, https://www.researchgate.net/publication/371216905_Phytochemicals_of_Hibiscus_sabdariffa_with_Therapeutic_Potential_against_SARS-CoV-2_A_Molecular_Docking_Study
21. Design of non-covalent dual-acting inhibitors for proteases MPRO and PLPRO of coronavirus SARS-CoV-2 through evolutionary library generation, pharmacophore profile matching, and molecular docking calculations - ResearchGate, acceso: abril 26, 2026, https://www.researchgate.net/publication/387599448_Design_of_non-covalent_dual-acting_inhibitors_for_proteases_MPRO_and_PLPRO_of_coronavirus_SARS-CoV-2_through_evolutionary_library_generation_pharmacophore_profile_matching_and_molecular_docking_calcul
22. Discovery of new drug indications for COVID-19: A drug repurposing approach | PLOS One, acceso: abril 26, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0267095
23. Predicting In Vitro and In Vivo Anti-SARS-CoV-2 Activities of Antivirals by Intracellular Bioavailability and Biochemical Activity | ACS Omega - ACS

Publications, acceso: abril 26, 2026, https://pubs.acs.org/doi/10.1021/acsomega.2c05376
24. Small molecules targeting endolysosomal acidification and signaling in sepsis and severe SARS-CoV-2 infection/COVID-19 - Open Exploration, acceso: abril 26, 2026, https://www.explorationpub.com/Journals/ei/Article/100363
25. Postacute Coronavirus (COVID-19) Syndrome - StatPearls - NCBI Bookshelf - NIH, acceso: abril 26, 2026, https://www.ncbi.nlm.nih.gov/books/NBK570608/
26. JAK inhibitors remove innate immune barriers to allow viral ... - bioRxiv, acceso: abril 26, 2026, https://www.biorxiv.org/content/10.1101/2025.02.17.638720v1.full
27. (PDF) Influenza Virus Down-Modulates G6PD Expression and Activity to Induce Oxidative Stress and Promote Its Replication - ResearchGate, acceso: abril 26, 2026, https://www.researchgate.net/publication/357673533_Influenza_Virus_Down-Modulates_G6PD_Expression_and_Activity_to_Induce_Oxidative_Stress_and_Promote_Its_Replication
28. The Role of Platelet Molecules in Risk Stratification of Patients with COVID-19 - MDPI, acceso: abril 26, 2026, https://www.mdpi.com/2673-6357/4/4/29
29. A dissection of SARS-CoV2 with clinical implications (Review) - Spandidos Publications, acceso: abril 26, 2026, https://www.spandidos-publications.com/10.3892/ijmm.2020.4636
30. Full article: A KSHV-targeted small molecule efficiently blocks SARS-CoV-2 infection via inhibiting expression of EGFR and Cyclin A2 - Taylor & Francis, acceso: abril 26, 2026, https://www.tandfonline.com/doi/full/10.1080/22221751.2024.2440490
31. Herbal medicine (Curcuma longa) inspired volatile compounds as potential inhalants against SARS-CoV-2: An in-silico strategy and modeling study - ResearchGate, acceso: abril 26, 2026, https://www.researchgate.net/publication/369953268_Herbal_medicine_Curcuma_longa_inspired_volatile_compounds_as_potential_inhalants_against_SARS-CoV-2_An_in-silico_strategy_and_modeling_study
32. Correlation of Viral Load With the Clinical and Biochemical Profiles of COVID-19 Patients, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8388060/
33. Combinatorial analysis reveals highly coordinated early-stage immune reactions that predict later antiviral immunity in mild COVID-19 patients - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8960124/
34. COVID-19 Hijacking of the Host Epigenome: Mechanisms, Biomarkers and Long-Term Consequences - PMC, acceso: abril 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12607690/
35. Cellular and Molecular Pathways of COVID-19 and Potential Points of Therapeutic Intervention - Frontiers, acceso: abril 26, 2026, https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2020.01169/full